Options Concentrations as Exit Information

Target placement, causal snapshot clocks, and native-index level tests

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Abstract

Can options concentrations improve a trade without predicting its entry? We synthesize completed Nasdaq futures target-intervention research and native-NDX level tests. Across 275 admitted sessions, same-day Gamma and Delta target caps produce resolved net subtotals of $31,706 and $31,431 on 421 fills. On 268 common known dates, their increments over unchanged entries and management are $4,590 and $4,315. A stricter 246-date comparison retains positive increments against both a half-distance target and a prior-session target-geometry distribution. All six primary adjusted confidence intervals nevertheless include zero, and recent windows favor the controls. Separate native-index contact tests find a modest retrospective improvement in ranking whether observed contact extremes remain final, but no improvement in fresh-reaction prediction. The useful research lead is therefore conditional exit placement, not a high-probability wall-fading rule. We distinguish delivered exposure estimates from dealer inventory, preserve snapshot and execution clocks, and report missing outcomes without turning them into zero returns. Economic evidence ends September 4, 2026; analytical level evidence ends September 18, 2026.

Keywords: options exposure; Nasdaq futures; target placement; market microstructure; causal availability; forecast evaluation.
JEL: G12, G13, G14.

1 The economic question

Options-derived levels can enter a trading decision in several distinct ways: where to initiate, which existing orders to avoid, or where to take profit. Those are different hypotheses. A concentration that helps identify an obstacle along a profitable path need not forecast an immediate reversal at first contact. Conversely, a visually compelling reaction near a strike is not evidence that an entry at that strike was available or profitable after costs.

This paper's main contribution is a controlled comparison of those roles, emphasizing the strongest supported positive lead in the existing research: *same-day options-informed target caps on an independently defined passive entry engine*. The contribution is not a newly optimized strategy. The original study evaluated 50 fixed alternatives; a subsequent 14-policy study added stronger target-geometry controls. We report the surrounding comparisons rather than relabeling the two favorable cells as an independently discovered optimum.

A second contribution is to separate economic usefulness from standalone level prediction. Native-NDX studies supply both an options-selected level population and an options-independent contact population. Together they show why a conditional exit result can coexist with weak evidence for fresh local reversals. This is retrospective research, not a recommendation to trade.

2 What an exposure surface measures

2.1 Greeks, quantities, and the missing inventory sign

For option i, let $V_i(S, \sigma, \tau)$ be its value and define

$$\Delta_i = \frac{\partial V_i}{\partial S}, \qquad \Gamma_i = \frac{\partial^2 V_i}{\partial S^2}, \qquad \mathcal{V}_i = \frac{\partial \Delta_i}{\partial \sigma}, \qquad \mathcal{C}_i = \frac{\partial \Delta_i}{\partial t}. \tag{1}$$

Vanna and charm conventions, scaling, and time signs must follow the provider definition. These sensitivities describe a pricing model, not a position census. In a stylized delta hedge, a dealer holding signed quantities q_i would require approximately

$$dH \simeq -\sum_i q_i m_i \left(\Gamma_i \, dS + \mathcal{V}_i \, d\sigma + \mathcal{C}_i \, dt\right), \tag{2}$$

where m_i is the contract multiplier. Equation 2 is a mechanism illustration, not an estimated flow equation in this paper. The data do not identify the q_i, beneficial owners, opening versus closing trades, complete dealer hedges, or the impact coefficient translating hedging into prices.

The classic pricing framework motivates sensitivities [1]; demand-based option pricing and expiration-date clustering motivate examining concentrations [2, 3]. Neither literature turns a vendor exposure field into measured dealer inventory. Here, a local level is selected from *delivered, model-based exposure*. Gross concentration, signed delivered exposure, open interest, executed option volume, and aggressive futures flow are not interchangeable. Gamma and Delta are evaluated separately, never added in incompatible units. Calendar same-day expiry is not certification of AM/PM settlement.

2.2 Information clocks and coordinate systems

For the economic intervention, snapshots are requested at 09:45, 10:00, 11:29, 12:59, and 14:29 America/New_York. Modeled availability is request time plus 60 seconds, with a 90-minute maximum age. The latest scheduled available tier owns a decision even if its data are missing: an older successful response cannot silently replace it. Historical first-seen and publication times remain unknown. A later download receipt does not establish historical tradability.

For a strike K and decision t, the retained basis proxy is

$$\widehat{L}_t^{NQ}(K) = K + F_{u(t)} - S_{u(t)}^{NDX}, \qquad u(t) = \text{same completed minute available before the decision.} \tag{3}$$

F uses the validated research-engine instrument identity. This is an approximate historical basis translation, not synchronous certified mapping or an orderable contract identifier. No current futures contract is substituted for a historical one.

The native-NDX analytical studies instead keep both contacts and levels in index points. They do not need Equation 3 to label a reaction. Their 09:45 and 10:00 surfaces are conditionally available at 09:46 and 10:01. The contact study additionally defines previous-close and 09:30 tiers,

retaining missing tiers explicitly; historical 09:30 surfaces are absent. Price features may use the completed contact candle, but options and developing-profile features stop at its *open*. Thus the close establishing a contact cannot retroactively determine its pre-existing options context.

3 The fixed economic experiment

3.1 Entry engine and conservative management

The baseline is a *passive EMA-center mean-reversion strategy with delayed break-even protection and three fixed intraday order blocks*. Table 1 defines its features and scheduling from the frozen implementation. Let t index completed one-minute bars; C_t, H_t, L_t, v_t denote close, high, low, and volume, and A_t the latest available *unrounded* five-minute Wilder ATR14. Let c be EMA21 rounded half-up to the nearest 0.25-point tick and R the tick-ceiling of A_t, at least one tick. Its two OCO legs are

$$e_d = c - dR, \quad s_d = e_d - dR, \quad T_0 = c, \qquad d \in \{+1, -1\}. \tag{4}$$

One NQ is used, with \$20 per point. The three America/New_York blocks are [10:00,11:30), [11:30,13:00), and [13:00,15:00). A limit needs one-tick penetration; both-side penetration in the same second cancels the OCO rather than awarding a favorable fill.

Stage	Frozen definition and decision rule
Price references	Session-anchored $W_t = \sum_{u \le t} v_u(H_u + L_u + C_u)/[3\sum_{u \le t} v_u]$, starting at 09:30. $E_t^{(j)}$ is the session-local close EMA, $j \in \{9, 21, 50\}$, smoothing $2/(j+1)$, `adjust=False`, minimum j observations, `ignore_na=False`. Missing minute slots remain in the grid.
ATR availability	Wilder ATR14 uses completed five-minute bars containing all five observed minute bars, preserving the original month-seeded history. A_t uses only an ATR available by t; finite EMA21 and finite positive A_t are required.
Trend gates	$s_{k,t} = \lvert W_t - W_{t-k}\rvert/A_t$, for $k = 5, 15$; require $s_{5,t} \le 0.10$ and $s_{15,t} \le 0.15$. These are VWAP changes, not regression slopes, and are not divided by k.
Cloud gate	$w_t = [\max_j E_t^{(j)} - \min_j E_t^{(j)}]/A_t \le 0.30$, over currently available EMA spans (missing spans are skipped). The denominator is unrounded ATR, not tick-rounded order risk R.
Efficiency gate	$\mathrm{ER}_{12,t} = \lvert C_t - C_{t-12}\rvert / \sum_{j=1}^{12} \lvert C_{t-j+1} - C_{t-j}\rvert \le 0.30$. A fully observed flat path has ER = 0; missing required history does not pass the gate.
Order scheduler	Scan qualifying completed-minute decisions chronologically from 10:00. Place only when the original slot is free. Expiry is $\min(t + 180\ \mathrm{s}, \text{block end})$, exclusive; a no-fill frees the slot at expiry, a dual collision at its event time plus 1 ns. Later qualifying minutes may retry.
Capacity release	The first original fill consumes its block; maximum three fills/day, no daily-profit lock and no added cooldown. The next slot opens only after the original fixed-control exit plus 1 ns. An unknown original lifecycle halts the day. Filtering or changing an exit never creates replacement capacity.

Table 1: Compact passive-entry specification. All four gates are inclusive and jointly required. The reported target comparisons preserve this original scheduling path; they do not reschedule on altered exit outcomes. Manuscript source, aggregate results, and vector figures accompany the data and materials release [7].

Initial risk remains $1R$. A favorable $0.50R$ excursion requests a stop at entry plus one directional tick, effective only on the next observed second. The fill second receives no management benefit; the trigger second retains original protection. Later collisions resolve stop first, and adverse stop gaps take the worse opening price. The time exit is the first observed close from fill plus 1,800 seconds through an inclusive 120-second tolerance; absent evidence leaves the outcome unknown.

Both option-conditioned OCO decisions precede the original first penetration, which cancels both legs even when a filter rejects the winning leg. Original fixed-control exits reserve capacity: an earlier target or rejected trade cannot create another order. These are *paired exit interventions on frozen order slots*, not a newly rearbitrated own-capacity portfolio. Longer alternative exit lifecycles may overlap later frozen slots; reported sums do not establish a standalone one-position implementation.

3.2 Three options roles and two stronger controls

The option selector treats Gamma and Delta separately. For one Greek g, expiry group D, and strike K, let $e_{g,\tau,K,\ell}$ denote each observed delivered call/put exposure. Ranking uses *gross mass* $G_{g,D}(K) = \sum_{\tau \in D,\, \ell \text{ observed}} |e_{g,\tau,K,\ell}|$, not the signed sum $B_{g,D}(K) = \sum e_{g,\tau,K,\ell}$ or its absolute value. Thus opposite delivered signs do not cancel in G. Only observed legs contribute; an entirely unobserved strike/group is unknown, not zero. Partial delivered support is not a complete-chain position estimate. D separates same-day expiry from calendar days 1–7 within an inclusive $\pm 1.5\%$ snapshot-spot radius.

A peak must have positive mass, be no smaller than each adjacent union-support neighbor, and strictly exceed at least one. Missing neighbors cannot be bridged. Equal-mass plateaus collapse to their smallest strike; an endpoint may use its sole neighbor, but a singleton or entirely flat profile has no peak. Peaks rank by descending gross mass, then smaller reference and smaller anchor (both equal the strike for these wall selectors). In that order, admit at most three strictly above and three strictly below spot, exclude at-spot peaks, and require at least ten native-index points from *every* previously selected level in that selector. This separation is not imposed across different Greeks or families.

Location accepts only entries within ten NQ points of a mapped reference. Obstacle rejects a reference more than one tick ahead but before the original target. Target caps use the nearest qualifying ahead reference, conservatively tick-rounded and front-run by one tick, never extending T_0.

Writing $q = 0.25$, $n_0 = d(T_0 - e_d)/q$, and n_L for the admissible, front-run directional distance in ticks, the target rule is

$$T_{\text{cap}} = e_d + dq \min(n_0, n_L), \qquad n_L \geq 1. \tag{5}$$

If no level qualifies, T_0 is unchanged. Missing reference support is unknown, not "no level." The half-distance comparator uses $\max(1, \lfloor n_0/2 \rfloor)$. Each historical comparator draws $\rho = n_{\text{cap}}/n_0$ from *all known submitted legs*, filled or not, in the previous 20 admitted sessions, matched by Greek, side, and scheduled tier, then applies $\max(1, \lfloor n_0 \rho \rfloor)$. Five fixed replicas are retained. Current-day support is required; current coordinates and outcomes do not enter donor selection.

4 Where the positive economics occur

The original comparison spans August 7, 2025–September 4, 2026: 275 passive sessions and 4,262 submitted opportunities, including unfilled attempts. It tests 24 interventions per engine plus two baselines, not a fills-only sample. A separate lower-frequency morning engine (Powell severe-regime TP36/SL24, no break-even) supplies a transfer comparison. The two engines and all arms are alternatives, never an additive sleeve.

On the same 268 complete-known dates, the baseline contributes \$27,116 and 421 fills. Gamma caps increase that matched sum by \$4,590; Delta caps increase it by \$4,315. Their shifted-location

Policy	n	n/session	Win %	EV ($)	PF	Net* ($)	U
Baseline	428	1.56	69.16	64.43	1.305	27,578	0
Half-distance	428	1.56	69.86	56.41	1.267	24,143	0
Gamma target	421	1.53	73.63	75.31	1.425	31,706	7
Delta target	421	1.53	72.92	74.66	1.404	31,431	7

Table 2: Passive target-study results at $4 total cost per resolved round turn. n: resolved fills; U: unknown sessions; PF: profit factor. Net* is a resolved subtotal, not complete-period PnL when U>0. The half-distance row comes from the follow-up; unchanged baseline and real arms reproduce the original study.

controls contribute $22,091 and $26,716, respectively. Both real arms preserve every known baseline fill; 421/428, or 98.36%, retention over all dates reflects unknown options support, not intentional frequency reduction.

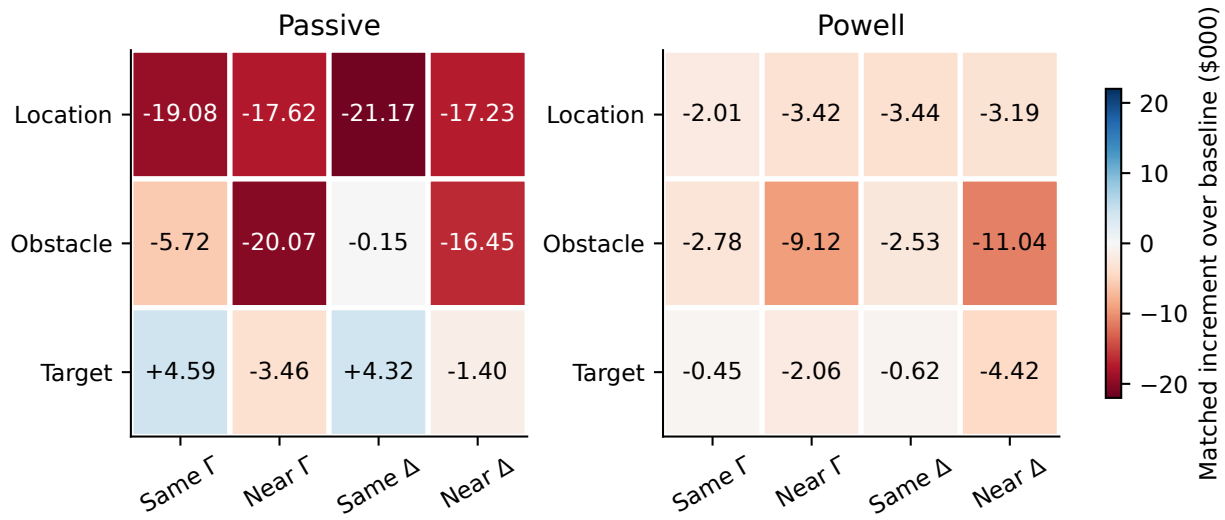


Figure 1: All 24 real intervention increments over their engine baseline, on each comparison's common-known dates. Same-day passive target caps are the two positive FULL cells. The figure reports matched dollars, not significance or complete-period returns; near-dated, location, and cross-engine results prevent a general "options always help" interpretation.

The economics have a recognizable tradeoff. Gamma's average net winner is $342.89 and average loser is −$671.97, compared with $398.91 and −$685.59 for the baseline. Delta averages $355.74 and −$682.29. Smaller average wins accompany more winning exits, not bounded or negligible downside. A roughly 73% trade win rate is not a probability that an arbitrary options-wall contact reverses.

A distinct Delta obstacle policy retains 219 resolved fills (0.80 per admitted session), yielding $26,964, PF 1.680, and $123.12 EV per fill. It forgoes 202 known baseline fills contributing only $152 net at the primary cost. This is a useful efficiency observation, but not an improvement in primary-cost total earnings, and not a second component to add to the target strategy.

5 Does any nearer target explain the result?

The follow-up compares baseline, half-distance, real Gamma/Delta caps, and five historical replicas for each Greek. For each Greek, all eight relevant arms must have a complete-known day. This leaves 246 of 275 admitted dates and 384 resolved fills per arm. On that common support, baseline and half-distance sums are \$24,924 and \$24,879; Gamma real is \$31,264 and Delta real \$32,109. These are matched subset sums, not all-date profits. In particular, the Delta subset exceeds its \$31,431 full resolved subtotal because excluded dates include losses.

Greek	H0	H1	H2	H3	H4	Mean	Real
Gamma	25,224	33,029	19,229	15,179	24,909	23,514	31,264
Delta	22,289	24,879	15,079	19,604	23,119	20,994	32,109

Table 3: All five historical target-geometry replicas on common-known FULL support, dollars at \$4 cost. The mean is a statistical comparator, not a sixth replica or combined trading policy.

Both real caps beat half-distance and the historical mean in EARLY and LATER. Gamma's increments over the historical mean are \$5,717 and \$2,033; Delta's are \$6,688 and \$4,427. EARLY comprises the first 183 admitted dates through April 28, 2026; LATER the remaining 92. Matched support is 158 and 88 dates. These are chronological stability descriptions on previously inspected history, not untouched validation.

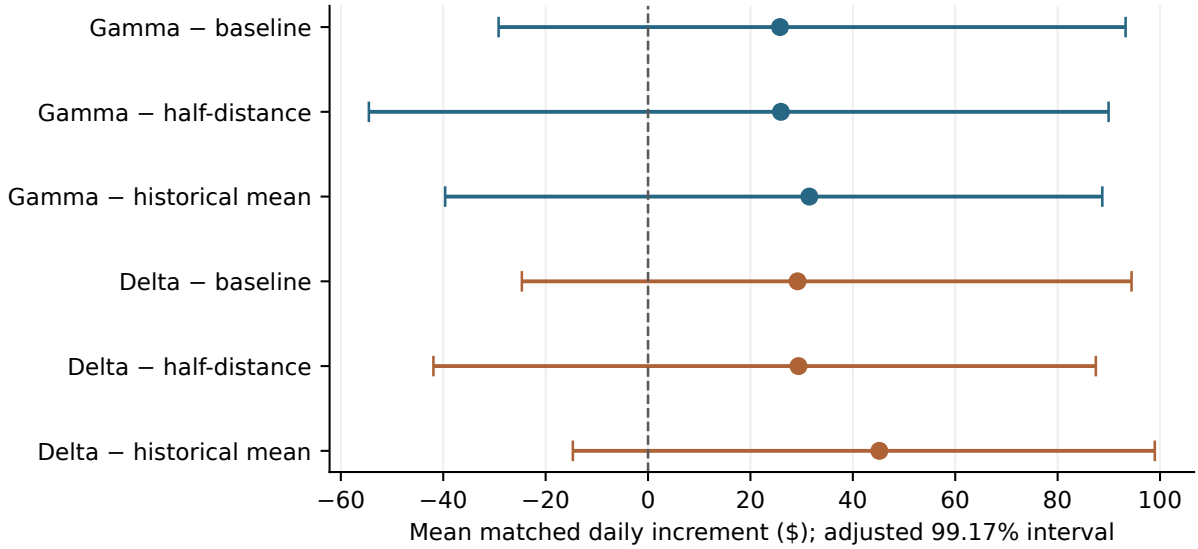


Figure 2: Target-policy increments on 246 common-known dates. Frozen 4,000-resample, five-admitted-session forward-block bootstrap; blocks never bridge unknown dates. Bonferroni correction across six primary contrasts gives 99.1667% intervals. Every interval includes zero. The horizontal axis is mean incremental dollars per matched day, not total profit.

For matched set $\mathcal{D}$, the estimand is

$$\widehat{\delta}_{a,b} = \frac{1}{|\mathcal{D}|}\sum_{s\in\mathcal{D}}(P_{a,s} - P_{b,s}), \quad P_{H,s} = \frac{1}{5}\sum_{j=0}^{4} P_{H_j,s}. \tag{6}$$

The full Gamma-minus-history mean is \$31.50/day, with adjusted interval [−\$39.62, \$88.73]; Delta

is \$45.18/day, [−\$14.69, \$98.98]. Bootstrap uncertainty is descriptive [4]; multiplicity correction here does not erase the earlier research search [5]. Gamma also loses to one FULL and two LATER historical replicas. Thus the controls weaken the explanation that *any* shortening works equally well, without isolating dealer information or establishing a reliable incremental edge.

6 Costs, missingness, and fixed recency

Arithmetic cost stress holds paths and fills fixed. For total round-turn charges \$4, \$14, \$24, and \$26.50, the Gamma resolved subtotal is \$31,706, \$27,496, \$23,286, and \$22,233.50; Delta is \$31,431, \$27,221, \$23,011, and \$21,958.50. These remain positive on resolved support, but identical fill counts imply

$$P_a(c) = P_a(4) - (c-4)n_a, \quad [P_a(c) - P_b(c)] = [P_a(4) - P_b(4)] \quad \text{if } n_a = n_b. \tag{7}$$

Invariant increments under a common charge are therefore arithmetic, not independent evidence about actual queue position, market impact, or slippage.

	Policy	Src	n	C	Net (\$)	EV (\$)	PF	DD (\$)	G/R/Z/U
L5	Baseline	51	6	0	956	159.33	2.92	498	3/1/1/0
L5	Half-distance	51	6	0	1,266	211.00	3.54	123	3/1/1/0
L5	Gamma target	51	6	0	731	121.83	2.46	498	3/1/1/0
L5	Delta target	51	6	0	956	159.33	2.92	498	3/1/1/0
L10	Baseline	142	13	0	2,503	192.54	3.48	1,006	7/2/1/0
L10	Half-distance	142	13	0	2,613	201.00	3.59	509	7/2/1/0
L10	Gamma target	142	13	0	1,638	126.00	2.62	1,006	7/2/1/0
L10	Delta target	142	13	0	1,758	135.23	2.74	1,006	7/2/1/0
L30	Baseline	492	45	0	2,175	48.33	1.24	3,521	18/9/3/0
L30	Half-distance	492	45	0	4,100	91.11	1.45	2,579	18/9/3/0
L30	Gamma target	492	45	0	1,640	36.44	1.19	3,759	19/8/3/0
L30	Delta target	492	45	0	2,985	66.33	1.35	2,624	19/8/3/0

Table 4: Fixed trailing admitted-session economics ending September 4, 2026, at \$4 cost. Src: submitted source opportunities; n: accepted = resolved; C: censored; DD: maximum known-segment daily-close drawdown; G/R/Z/U: green/red/zero/unknown days. All shown windows are completely known, so net equals complete-known-day PnL and both EV denominators equal n. Exact dates and remaining support metrics appear in the appendix.

Recency does not confirm the full-history ranking. Gamma trails baseline by \$225, \$865, and \$535 in L5/L10/L30. Delta matches baseline in L5, trails by \$745 in L10, and leads by \$810 in L30. Both real caps trail half-distance and their historical mean in all three windows. These fixed windows are descriptive: there is no recent-performance switch, fitted hybrid, or favorable-window selection.

Full-history real-cap outcomes are unknown on September 1 and November 27, 2025, and January 19, February 16, May 25, June 19, and July 3, 2026. These cash-market holidays are not random missingness. The full-period real-cap PnL is therefore null. Historical controls additionally retain warmup and donor gaps, totaling 29 unknown sessions each. Resolved subtotals on an unknown day never become complete-day PnL.

Observed known-segment drawdowns are \$8,756 baseline, \$7,218 half-distance, \$6,630 Gamma, and \$6,030 Delta. Each unknown date resets the options drawdown path, so these are not comparable uninterrupted full-period risk guarantees. They also omit intraday drawdown. An equity curve that bridges the gaps would imply knowledge the study does not have; none is constructed here.

7 Do options-selected levels predict fresh reactions?

The options-native discovery study addresses a different question: which levels deserve attention before any passive trade exists? Across 281 admitted native-NDX sessions, August 7, 2025–September 18, 2026, it evaluates six frozen hypotheses and one shifted control for each. Four fade contact; low-exposure gaps and prominent-wall breaks follow continuation. The primary label is a fresh favorable 20 points before adverse 20, within at most 15 minutes after the decision close. The decision candle is excluded. Ambiguous and incomplete labels remain unknown; observed no-hit timeouts are non-successes in the known-outcome fraction.

Level family	Full n	n/session	Full %	Later %	Shifted %
Same-day peaks	1,090	3.88	50.1	52.9	47.6
Near-dated peaks	952	3.39	47.6	50.0	49.2
Expiry agreement	981	3.49	48.3	50.5	48.3
Cluster centers	1,031	3.67	49.7	50.3	53.0
Exposure gaps	298	1.06	50.0	49.6	52.7
Wall breaks	570	2.03	46.0	47.2	46.0

Table 5: Options-selected native-NDX response observations, not fills or achievable trade frequency. Later history is 94 sessions from May 6, 2026; shifted rates use their own supported populations. Counts overlap across hypotheses and must not be summed as a portfolio.

The most attractive raw later contrast is same-day peaks: 52.9% real versus 47.6% shifted. Yet the predeclared equal-session comparison on 77 jointly supported dates is approximately zero, with adjusted interval $[-13.4, +13.3]$ percentage points. Different contact populations and event weighting explain why a raw five-point advantage need not survive the paired estimand.

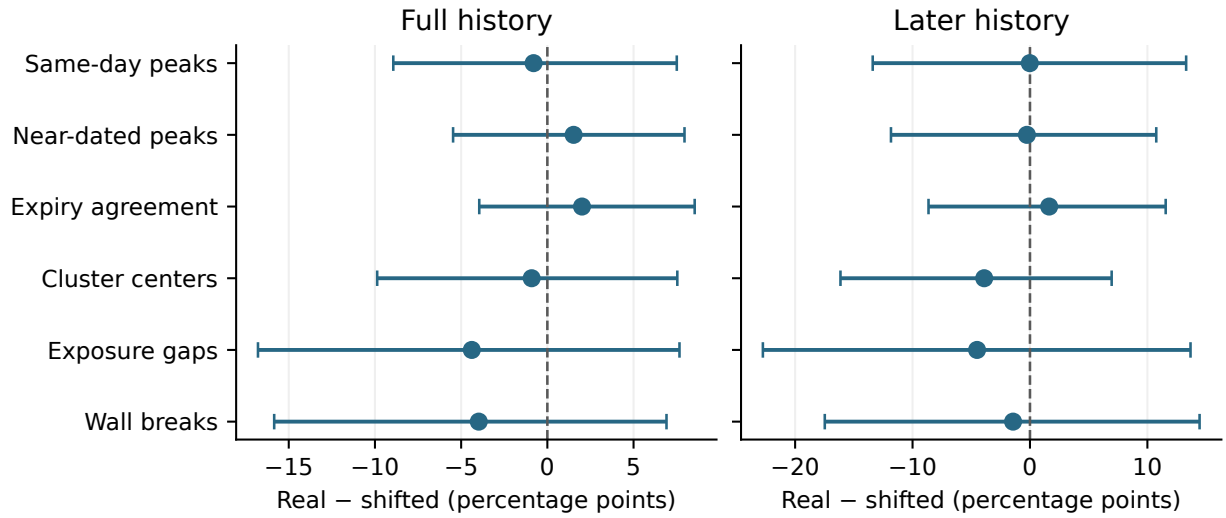


Figure 3: All six primary real-minus-shifted response contrasts. Points and multiplicity-adjusted intervals come directly from the completed native-index study. Both full and later panels retain every hypothesis; every interval crosses zero. These are response probabilities on supported contacts, not PnL or dealer-impact estimates.

Snapshot ownership matters for this denominator. An event tied to the 09:45 tier cannot have a complete no-hit 15-minute horizon before the tier's exclusive expiry. Expiry-related censoring means

known-outcome fractions are not unconditional reversal probabilities. Shift controls preserve strike support, missingness, and paired expiry weights, but not exact level density, distance, or contact time. None of the six standalone formulations establishes a high-probability level model. That result does not contradict the target intervention: a useful conditional exit may exploit the location of an obstacle without correctly predicting the direction of every subsequent contact.

8 Contacts, calibration, and final-extreme ranking

A separate contact-first study removes options from population selection. Price must approach a native-NDX 25-point grid's ±5-point band after arming from at least 15 points away. Recontacts require two qualifying subsequent closes and ten minutes' cooldown. There are 17,972 contacts, 15,002 eligible, with 7,292 fresh-reaction successes, 7,223 failures, and 487 unknown labels. Here the horizon is 30 minutes; neither barrier, incomplete horizons, or same-minute ambiguity are unknown. This is not the discovery study's timeout convention.

The *final-extreme* label is separate from that fresh-reaction label. Write $a = +1$ for an approach from below (the fade-short side) and $a = -1$ for an approach from above (fade-long). With complete, timely coverage of the full cash session, including any scheduled early close, a contact receives label one exactly when its candle is the *earliest tied full-session high* for $a = +1$, or the *earliest tied full-session low* for $a = -1$. Other contacts receive zero; incomplete full-session coverage leaves the label unknown. Thus the model asks whether the *already observed contact candle's extreme remains the final session extreme*; it does not predict the location or time of a future extreme. Later ties do not replace the earliest occurrence.

Options concentration is the empirical percentile of observed gross exposure within each Greek's delivered 25-point bins. Persistence requires current and fixed-predecessor same-bin top-three Gamma membership, positive exact matched-leg support, and unchanged support. Matched change intersects expiry/strike/leg identities. The profile is a native-index minute-range TPO approximation, not a trade-volume profile. Gamma context is known at only 6,841 of 15,002 eligible contacts (45.6%); missing predictors do not remove the contact.

The models are fixed L2 logistic regressions ($C = 0.1$), with train-only preprocessing, missing indicators, and equal-session weighting. PRICE_PROFILE contains 11 price/profile/support covariates; OPTIONS adds seven options features. SHIFTED_OPTIONS moves whole bin records. Fit ends April 30, 2026. The 92 later sessions through September 11 contain 6,255 known reaction labels and only 54 final-extreme positives among 6,395 known rows.

Model	Reac. Brier	Reac. AUC	Extr. Brier	Extr. AUC	Extr. AP
BASE_RATE	0.249911	0.5000	0.009176	0.5000	0.0093
PRICE_PROFILE	0.252852	0.4977	0.009774	0.8184	0.1004
OPTIONS	0.252919	0.4955	0.009578	0.8285	0.1098
SHIFTED_OPTIONS	0.253267	0.4962	0.009844	0.8089	0.1029

Table 6: Later, out-of-fit but previously inspected history. Brier scores are lower-is-better probability losses; AUC and average precision (AP) describe ranking. Equal-session weighting applies. No model or threshold is changed for this paper.

OPTIONS improves extreme AUC from 0.8184 to 0.8285 and AP from 0.1004 to 0.1098, a genuine but limited retrospective ranking lead. Its extreme Brier score is still worse than the constant baseline. At the frozen 0.55 threshold, session-weighted precision/recall are 23.8%/4.9%. Across the

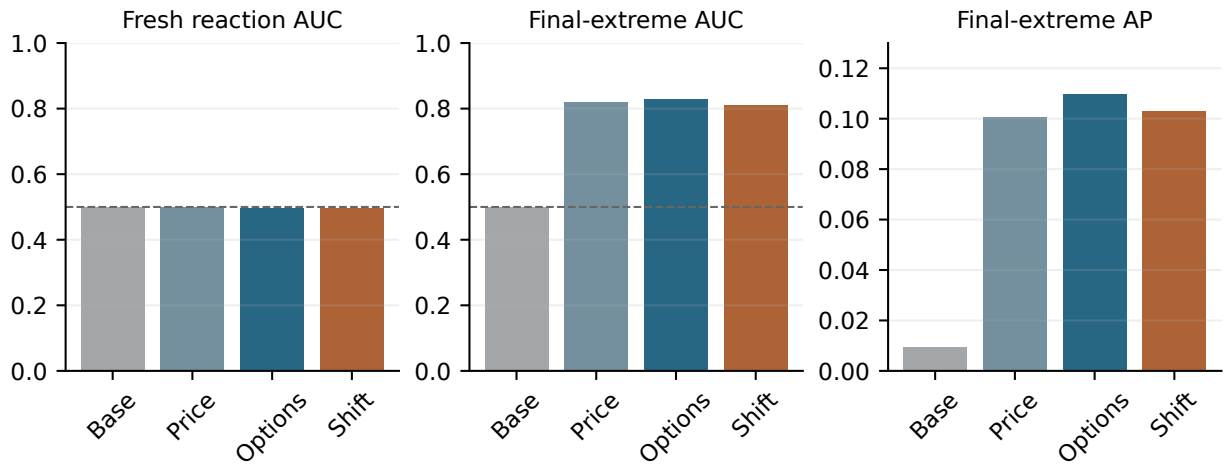


Figure 4: Prediction tasks must not be conflated. Fresh-reaction AUC stays near chance. Ranking whether the observed contact extreme remains final improves modestly with options, but AP remains 0.1098 and threshold recall remains low. Dashed AUC lines mark 0.5, not a profitability threshold.

full contact population, only 105 of 558 known daily extremes are even represented. Proper scoring rules distinguish probability quality from discrimination [6]; neither AUC nor a rare-event ranking improvement establishes a forecast of a future HOD/LOD location. The analytical studies' dollar economics are N/A in this paper, not zero and not borrowed from the target study.

9 Interpretation and research implications

The positive finding is economically concrete: on an existing frequent entry engine, same-day options target placement improved historical resolved earnings without deliberately filtering known fills. Stronger controls retained positive matched increments in both chronological panels. The effect is specific to *role, expiry, and engine*. Near-dated target caps, location restrictions, and the second entry engine did not show the same FULL improvement. A universal options-level reversal story is unnecessary to explain the observed target result and unsupported by the standalone contact evidence.

Three identification limits are material. First, demand and inventory are not directly measured. Current-day target geometry may capture price-state information correlated with options concentrations rather than dealer hedging. The history controls match prior relative geometry conditional on side and tier, not exact current distance or density. Second, conditional historical publication clocks and basis translation are weaker than observed, contemporaneous decision availability. Third, these are previously inspected samples: selecting this paper's positive lead from a broad research program is not independent validation. Adjusted intervals, unfavorable recency, and missing days therefore belong beside the positive dollars rather than in a separate disclaimer.

A defensible next empirical question is whether the *unchanged* target policies retain their incremental value with observed first-seen options availability and independently validated mapping. Reoptimizing targets or switching policies based on the displayed recency would answer a different question.

Conclusion. Options concentrations are most promising here as conditional information about where to exit, not as autonomous instructions about where price must turn. The target result is a substantive positive historical finding with explicit competing explanations. The native-index studies provide a complementary research asset: causal populations and falsification tests that stop attractive visual levels from being mistaken for measured dealer positions, calibrated reversal forecasts, or live trading performance.

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A Fixed-window membership and metric support

All dates below are from the original admitted economic calendar, not inferred weekdays. L5/L10/L30 end at the economic package's September 4 cutoff; the September 18 analytical cutoff does not extend these PnL windows. No window is chosen for its outcome.

L5 (2026-09-04 cutoff). 2026-08-31, 2026-09-01, 2026-09-02, 2026-09-03, 2026-09-04.

L10 (2026-09-04 cutoff). 2026-08-24, 2026-08-25, 2026-08-26, 2026-08-27, 2026-08-28, 2026-08-31, 2026-09-01, 2026-09-02, 2026-09-03, 2026-09-04.

L30 (2026-09-04 cutoff). 2026-07-27, 2026-07-28, 2026-07-29, 2026-07-30, 2026-07-31, 2026-08-03, 2026-08-04, 2026-08-05, 2026-08-06, 2026-08-07, 2026-08-10, 2026-08-11, 2026-08-12, 2026-08-13, 2026-08-14, 2026-08-17, 2026-08-18, 2026-08-19, 2026-08-20, 2026-08-21, 2026-08-24, 2026-08-25, 2026-08-26, 2026-08-27, 2026-08-28, 2026-08-31, 2026-09-01, 2026-09-02, 2026-09-03, 2026-09-04.

Policy		GP ($)	GL ($)	W/L/0	Win %	Best date	Best ($)	Share
Baseline	L5	1,455	499	5/1/0	83.3	09-03	926	0.969
Baseline	L10	3,511	1,008	11/2/0	84.6	09-03	926	0.370
Baseline	L30	11,312	9,137	32/13/0	71.1	08-12	1,322	0.608
Half-distance	L5	1,765	499	5/1/0	83.3	09-02	577	0.456
Half-distance	L10	3,621	1,008	11/2/0	84.6	08-25	692	0.265
Half-distance	L30	13,237	9,137	32/13/0	71.1	07-31	1,632	0.398
Gamma target	L5	1,230	499	5/1/0	83.3	09-02	527	0.721
Gamma target	L10	2,646	1,008	11/2/0	84.6	08-25	932	0.569
Gamma target	L30	10,288	8,648	33/12/0	73.3	08-06	1,448	0.883
Delta target	L5	1,455	499	5/1/0	83.3	09-03	926	0.969
Delta target	L10	2,766	1,008	11/2/0	84.6	09-03	926	0.527
Delta target	L30	11,633	8,648	33/12/0	73.3	08-06	1,448	0.485

Table 7: Remaining recency support for Table 4. GP/GL are sums of positive/negative net trade outcomes, not pre-cost cash flows. W/L/0 are resolved wins, losses, and zero-PnL trades; best dates are MM-DD in 2026. Best-day share divides the best complete-known day by complete-known-day net.

For all twelve displayed policy/window rows, accepted = resolved, censored = 0, unknown days = 0, and complete-known-day net = resolved net. Thus EV/accepted = EV/resolved. Win-rate support is W+L; day shares divide each G/R/Z/U count by the entire fixed window (5, 10, or 30), not by active days. A zero-trade known session stays in that denominator. Profit factor is GP/GL: a positive GP with zero GL is $+\infty$; no gains or losses is unsupported/null. No such edge case occurs in the displayed rows. A best-day share above one is possible when other days lose; it is not a stability score.

Reproduction and scope. The public data and materials release [7] accompanies this manuscript. Its aggregate supplement retains every target-control arm, all 50 original economic alternatives, all fixed cost scenarios, exact window membership, and standard economic metrics from the repository's existing fixed-cutoff engine. It retains full target-control daily aggregates and the original experiment's trailing daily aggregates, including null states, plus full unknown-date lists. It exports no quotes, trade timestamps, order identifiers, raw option surfaces, or private fill records. The figures are vector PDFs with PNG previews, generated directly from pinned public aggregate authorities. The machine-readable source index supplies precise evidence identities without using the manuscript as a hash ledger.

The economic sources are the completed *Options-conditioned execution* and *Options target controls* studies. The analytical sources are *Options-native discovery* and *Native-NDX options-level contacts*. Their family versions, calendars, and outcome definitions remain distinct.